

Multiple Choice (10 marks)

1. Which of the jovian planets have rings?
 - (a) Neptune
 - (b) Uranus
 - (c) Saturn
 - (d) all of the above
2. What element is most common among the Jovian Planets?
 - (a) Hydrogen
 - (b) Helium
 - (c) Carbon
 - (d) Oxygen
3. According to the Solar Nebular theory what are asteroids and comets?
 - (a) They are the shattered remains of collisions between planets.
 - (b) They are chunks of rock or ice that condensed long after the planets and moons had formed.
 - (c) They are chunks of rock or ice that were expelled from planets by volcanoes.
 - (d) They are leftover planetesimals that never accreted into planets.
4. Where is the crater from the impact that is believed to be responsible for the mass extinction of dinosaurs 65 million years ago?
 - (a) Tunguska Siberia.
 - (b) Chicxulub Crater Yucatan Peninsula in Mexico.
 - (c) Quebec Canada.
 - (d) Meteor Crater in Arizona.
5. Which statement about an atom is not true:
 - (a) The nucleus contains most of the atom's mass but almost none of its volume.
 - (b) A neutral atom always has equal numbers of electrons and protons.
 - (c) A neutral atom always has equal numbers of neutrons and protons.
 - (d) The electrons can only orbit at particular energy levels.

True / False (10 marks)

Answer True or False

6. True or False: The nebular theory does not predict an equal number of terrestrial and jovian planets in the solar system.
7. True or False: Dione is one of the Galilean moons, recognized for its geological activity and proximity to Jupiter.
8. True or False: The ratio of solar radiation flux on Mercury's surface at perihelion to aphelion is 9:4.
9. True or False: All fresh impact craters formed on solid surfaces exhibit ejecta and raised rims as common features.
10. True or False: The presence of an atmosphere is the most important factor in determining a planet's volcanic and tectonic history.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the definition of a parsec in astronomy and analyze its significance in measuring astronomical distances, particularly in relation to the concept of arcseconds.
12. Discuss the processes by which the nuclei of comets contribute to meteor showers when Earth intersects their orbital paths, and analyze the implications for our understanding of these phenomena.

End of Paper